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## **Harnessing Ambient Heat for Sustainable and Efficient Underground Energy Storage**

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### **Abstract**

This study presents an innovative energy storage system utilizing subterranean flexible vessels to store energy in a form of compressed gas while leveraging ambient desert heat for thermal enhancement. The approach optimizes surrounding pressure and thermal retention, ensuring a cost-effective alternative to conventional storage methods. The paper shares the Thermodynamic model that would be utilized for the performance evaluation of the system; particularly its integration with renewable energy sources such as solar to enhance grid stability in high-temperature regions.

The system consists of underground flexible vessels connected to a gas supplier and a turbine generator. During surplus energy periods, gas is compressed and stored underground in the form of potential energy, passively absorbing heat from surrounding sand. When energy demand rises, the heated stored gas expands the flexible vessels which are connected via a turbine to generate electricity, completing a closed-loop cycle. A heat exchanger regulates gas temperature before compression, improving efficiency. This closed loop cycle demonstrates the nature of gas expansion, and heat transfer impact on energy output.

Extensive thermodynamic analysis confirm that this system achieves superior energy retention with minimal efficiency losses compared to conventional Compressed Air Energy Storage (CAES). Unlike battery-based alternatives, it reduces reliance on critical materials, lowers environmental impact, and enhances operational longevity. The ability to absorb ambient heat optimizes gas expansion efficiency, improving energy retrieval rates and overall performance. Thermal energy storage, heat recovery and the drive to enable near-isothermal operation has motivated this alternative design configurations; ultimately resulted in a U.S. granted patent: US 11,927,131 B1. By addressing challenges in energy security and long-term storage, this system presents a viable alternative for desert and arid regions. Field demonstrations will refine system performance and validate real-world efficiency gains.

This engineering approach introduces a scalable energy storage concept that utilizes the surrounding environmental pressure and heat to enhance thermodynamic efficiency. Unlike conventional systems requiring extensive cooling, this approach naturally optimizes performance while minimizing material consumption. This may serve as a foundation for contributing to next-generation energy storage solutions, enhancing sustainability and reducing dependence on fossil fuels.

## Introduction

The need for reliable and flexible energy storage solutions has become more urgent, especially in regions where environmental conditions present unique challenges. This is driven by the world moves steadily toward cleaner and more sustainable energy systems as discussed in [IEA 2024](#). In desert environments, where sunlight is abundant but energy demand can vary significantly between seasons, the pressure to find practical and scalable storage options is even greater. During summer, based on ([Al-Maktoumi, 2017](#)), electricity consumption often doubles compared to winter, placing a significant burden on supply systems and pushing the limits of conventional storage technologies.

[Zhang et al, 2019](#), presented a thermodynamic analysis of a novel Compressed Air Energy Storage (CAES) system aimed at improving round-trip efficiency. The proposed system incorporates thermal energy storage to recover waste heat generated during the compression phase, enabling near-isothermal operation. This research motivated the exploration of alternative design configurations, ultimately leading to a U.S. granted patent: US 11,927,131 B1:2024.

The patent introduces an innovative energy storage concept specifically designed for operation in arid and desert environments. The system stores energy in the form of compressed gas, such as carbon dioxide within flexible, subterranean vessels. A distinctive feature of this approach is its ability to passively absorb ambient heat from the surrounding desert environment during storage, thereby increasing the internal energy of the compressed gas over time. When electricity is required, the heated gas is released and expanded through a turbine to generate power based on anticipated demand. This integration of thermal enhancement with underground storage combined with process control loops offers a promising pathway for efficient, scalable, and climate-adapted energy storage.

This energy storage system is specifically engineered to operate in harmony with desert environments and addresses the conventional battery systems that are susceptible to degradations from extreme temperatures and ambient environments. [Hasan et al, 2021](#), has conducted complete assessment on the environmental conditions as well as operation and maintenance impacts related to solar PV cell and underlining infrastructure. Several factors were assessed that have an affect the economy of power generation This includes manufacturing material, installation, operation and maintenance, and environmental factors. Hence, this alternate energy generation option may be leveraged in desert region's natural heat, abundant open space, and geological stability to enhance performance. Moreover, the system considered in this paper capitalizes on the innovative energy storage system and the effect of an optimum output efficiency that is heavily impacted by the energy conversion. The introduced system's modularity and scalable design enables deployment in remote or off-grid locations where traditional energy storage solutions are impractical or economically unfeasible.

In this paper, we present the granted patent, US 11,927,131 B1:2024, scientific principles, integrated components, and operational capabilities of the system. We also explore its potential to support broader sustainability objectives, such as integrating renewable energy sources, enabling carbon reuse, and reducing dependence on fossil fuels for domestic power generation. This technology offers an alternative promising pathway toward a more resilient, efficient, and sustainable energy future.

## Method and Procedure

Energy, in its many forms; mechanical, thermal, chemical, electrical, and potential, plays a foundational role in powering human civilization. Among these, electrical energy stands out as the most versatile and widely used form in modern society. It drives everything from household appliances and industrial machinery to digital infrastructure and transportation systems. However, while the generation and utilization of electricity have advanced significantly, the challenge of storing electrical energy efficiently remains a critical bottleneck in the global energy landscape. This challenge was amplified by [International Energy Agency \(IEA\) 2023](#) report as it highlighted generation of electricity from renewable sources has become

increasingly cost-effective, the ability to store that electricity efficiently, affordably, and at scale remains a critical technical and economic challenge.

In He, Wei. et al, 2021, it was highlighted direct storage of electricity is not feasible due to the nature of electric charge, which must be either used immediately or converted into another form for storage. As a result, most energy storage systems today rely on intermediate forms, such as chemical energy stored in batteries. While batteries have become increasingly sophisticated, they come with limitations ranging from high costs and limited lifespans to environmental concerns related to resource extraction and disposal. These drawbacks have spurred a global effort among scientists and engineers to explore alternative, more sustainable, and scalable energy storage solutions.

This paper introduces a new methodology in energy storage that capitalizes on three elements; sand heat capacity, subterranean pressure attributes and the regulated process for gas replenishing. Figure 1 depicts the conceptual model for the proposed system design. The system contains two closed loops; one for the gas and the other for the coolant. The gas is processed through multiple stages that includes compression, cooling, and heating to ensure the energy storage is at its maximum efficiency. The coolant, on the other hand, is processed through two main stages; heating and storing. The difference between the two loops is that the gas stores energy whereas the coolant is used to produce energy in low quantities.

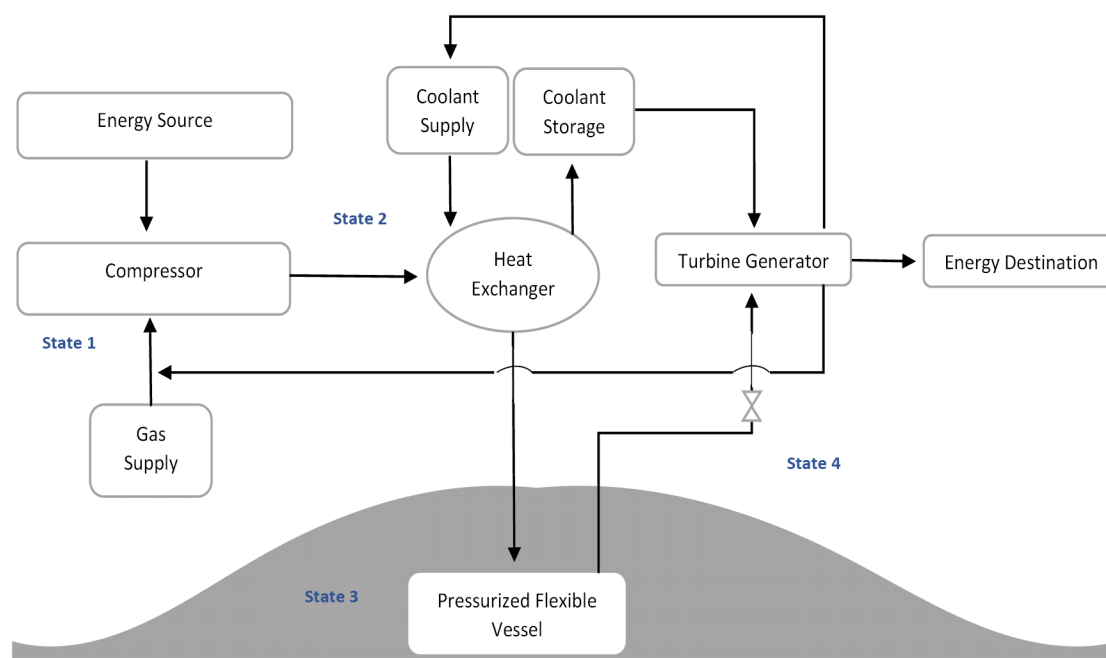


Figure 1—Underground Flexible-Vessel Compressed Gas Energy Storage System Integrated with Ambient Heat

Figure 2 below illustrates a dual-path energy recovery process in which compressed gas is cooled to yield both a heated coolant stream and cooled high-pressure gas. The heated coolant drives a turbine generator directly, while the cooled gas is stored, reheated from surrounding heat, and expanded through another turbine to generate electricity, with exhaust gas recycled into the system.

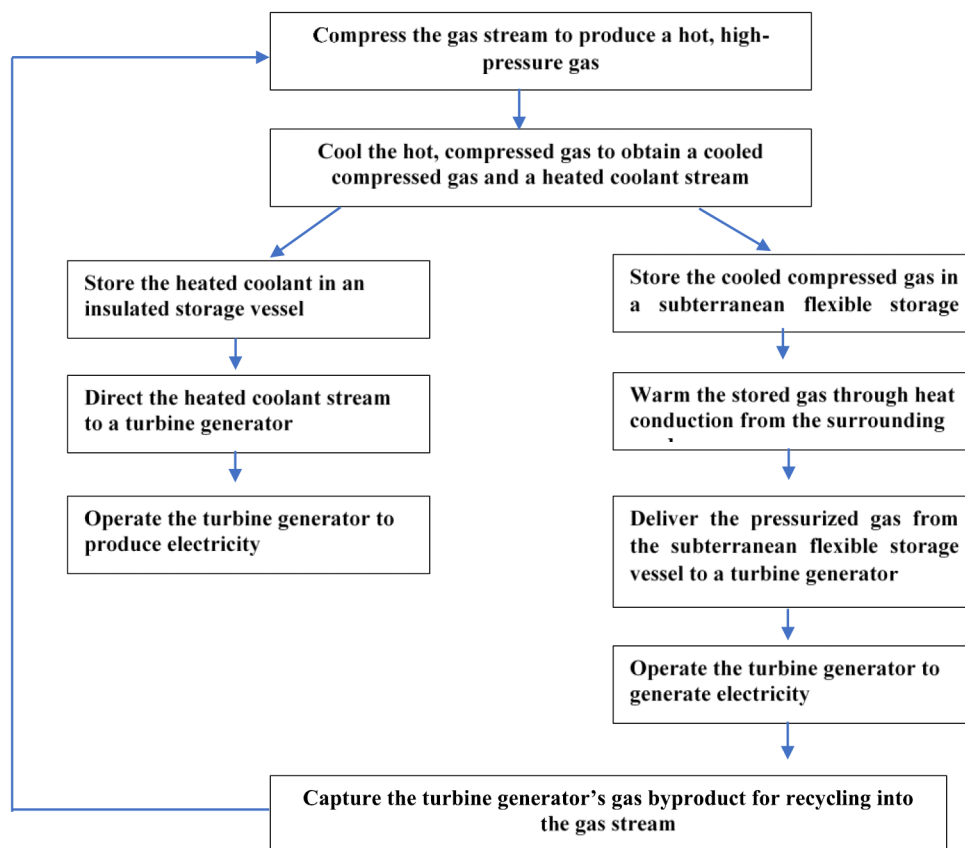


Figure 2—Process Flow Diagram

### Thermodynamic model

The mathematical model representation for the introduced infrastructure depends on thermodynamics principle where pressure and temperature are the main variables. To identify the impact of these variables, it is necessary to develop the intermediate impact for the control loops as they traverse from one state to the next. Table 1 illustrates the different states that control loops will be channeled through each one:

Table 1—State Flow Path

| States   | Path                                          |
|----------|-----------------------------------------------|
| State 1  | Gas Supply to Compressor (Inlet)              |
| State 2c | Compressor (Outlet)                           |
| State 2  | Compressor to Heat Exchanger                  |
| State 3  | Heat Exchanger to Pressurized Flexible Vessel |
| State 4  | Pressurized Flexible Vessel to Turbine        |

The charging phase (State 1) uses surplus energy to compress the working gas according to the isentropic compression equation.

1. Compression (State 1 to State 2c) drive the Isentropic temperature rise as show in the equation number 1; Isentropic temperature rise:

$$T_{2s} = T_1 \left( \frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}}$$

The temperature resulted from the Actual compressor outlet can be calculated in the equation number 2:

$$T_2 = T_1 + \frac{T_{2s} - T_1}{\eta_c}$$

The Compressor work can be calculated in equation number 3:

$$W_c = \frac{\frac{\gamma}{\gamma-1} RT_1 \left[ \left( \frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}} - 1 \right]}{\eta_c}$$

2. Cooler/Heat Exchanger (State 2c to State 2); during heat exchange phase (state 2), the compressed gas is being cooled by the coolant within the heat exchanger (HX):

$$Q_{HX} = c_p(T_{2c} - T_2)$$

3. Storage heating (State 2 to State 3) capitalizes on the heat transfer from the surrounding sand by increasing the gas internal energy

$$Q_{23} = c_p(T_3 - T_2)$$

4. Turbine Expansion (Step 3 to Step 4); in the discharging phase (state 4), expansion through the turbine follows:

Isentropic temperature drop

$$T_{4s} = T_3 \left( \frac{P_4}{P_3} \right)^{\frac{\gamma-1}{\gamma}}$$

Actual outlet temperature

$$T_4 = T_3 - \eta_t(T_3 - T_{4s})$$

Turbine work

$$W_t = \eta_t \frac{\gamma}{\gamma-1} RT_3 \left[ 1 - \left( \frac{P_4}{P_3} \right)^{\frac{\gamma-1}{\gamma}} \right]$$

5. Round trip efficiency;

$$\eta_{RT} = \frac{W_t}{W_c}$$

**Table 2—Thermodynamic Model Variables**

| State | P     | T        | Note                        |
|-------|-------|----------|-----------------------------|
| 1     | $P_1$ | $T_1$    | Compressor inlet            |
| 2c    | $P_2$ | $T_{2c}$ | After compressor            |
| 2     | $P_2$ | $T_2$    | After cooler (into storage) |
| 3     | $P_3$ | $T_3$    | After ambient heating       |
| 4     | $P_4$ | $T_4$    | Turbine exhaust             |

There are some energy losses as the gas travels from a phase to another except for the third phase where there is a gain in the energy due to the ambient heating received by the system. This compensates the losses caused by the compressor, heat exchanger, and the turbine generator making the system more efficient and long-lasting.

The proposed system adapts this principle but replaces gravitational potential energy with compressed gas as a form of stored potential energy. The core idea involves using flexible, gas-tight vessels buried deep underground in arid desert environments. During periods of excess electricity generation such as when solar or wind energy production exceeds demand the surplus electricity is used to compress gas (likely air or another inert gas) and store it under high pressure within these subterranean vessels.

One of the key innovations of this system lies in its integration with the natural thermal properties of desert environments. [Campanale et al. \(2025\)](#), delved into the thermal behavior of various types of sand sourced. The experimentation and analysis revealed a strong correlation between sand color, solar reflectance, and temperature variations have direct impact on the thermal properties of sand. The approach is to capitalize on the sand and earth surrounding the vessels to act as a thermal reservoir, allowing the compressed gas to absorb ambient heat. This heat absorption helps maintain the pressure of the gas, which is crucial for efficient energy recovery. When electricity is needed, the pressurized gas is released and directed through a turbine generator. The expansion of the heated gas through the turbine drives the generator, producing electricity. After passing through the turbine, the cooled, low-pressure gas is recirculated back to the gas supply unit to be recompressed and stored again, thus creating a closed-loop system.

To enhance the efficiency of the process, a heat exchanger is integrated into the system. This component ensures that the gas entering the compression stage is cooled to optimal levels before being stored, maximizing the energy density and minimizing losses during the compression phase.

The modular nature of this system allows for scalability. Multiple vessels can be interconnected with multiple turbine generators, enabling the creation of large-scale underground energy storage plants. This scalability makes the system particularly well-suited for regions with vast, uninhabited desert areas—such as those found across the Middle East. These regions not only have abundant renewable energy potential (especially solar), but also the geographic space and environmental conditions ideal for deploying such underground storage systems.

Moreover, because the system relies on natural thermal gradients and subterranean infrastructure, it has minimal visual and ecological impact compared to surface-level energy installations. This makes it a promising candidate for sustainable, large-scale energy storage that complements intermittent renewable energy sources like solar and wind.

## Comparison with Conventional CAES Systems

Compressed Air Energy Storage (CAES) has long been recognized as a viable grid-scale energy storage solution, especially for its ability to provide long-duration and high-power output. Traditional CAES systems typically store compressed air in large underground caverns or aquifers and recover the stored energy through expansion turbines. These systems are generally reliant on mechanical compressors powered by grid electricity and require significant capital investment for geological surveys, site development, and infrastructure to ensure cavern stability and safety.

One of the key differentiators of the proposed system is the use of modular high-strength gas vessels buried underground, as opposed to relying on natural caverns. This modularity not only enhances geographic flexibility—allowing deployment in regions without suitable geological formations—but also simplifies site preparation and permits phased scaling based on demand growth.

Thermally, traditional CAES often requires either fossil-fuel combustion (as in diabatic systems) or costly regenerative heat storage (in adiabatic systems) to mitigate heat loss during compression. In contrast, the proposed system innovatively leverages ambient desert heat to support thermal recharging, eliminating the need for additional fuel sources or active heat storage units. This significantly reduces both environmental impact and operational complexity ([Cardenas et al., 2025](#)).

From an efficiency standpoint, while traditional CAES systems exhibit round-trip efficiencies ranging between 40% and 55%, the integration of ambient heat in our system enhances thermal stability, which

can improve overall thermodynamic efficiency. Additionally, the closed-loop design of the vessel-based storage minimizes air leakage and pressure variability, issues commonly associated with large cavern-based systems (Chen et al., 2016).

The infrastructure footprint is also considerably lower. Traditional CAES installations often demand extensive above-ground mechanical and control systems, whereas the buried vessel approach maintains landscape integrity and visual aesthetics, particularly crucial for deployments in ecologically or culturally sensitive desert areas.

Economically, the modular design aligns better with decentralized energy systems, especially in remote or developing regions. This stands in contrast to conventional CAES facilities, which often require centralized grid infrastructure. Our solution supports distributed storage, enhances local energy resilience, and can be integrated seamlessly with off-grid renewable sources such as solar PV and thermal concentrators (Franco, 2025).

In summary, while traditional CAES remains an important part of the energy storage landscape, the presented approach overcomes many of its site, thermal, and operational limitations. By integrating ambient heat, vessel modularity, and minimal land impact, the proposed system represents a more scalable, sustainable, and regionally adaptable evolution of compressed air energy storage.

## Discussion

The introduced innovation in this paper composes of different components that are inter-locked via two closed loops; one for the gas and the other for the coolant. This section examines the two loops and their components in more detail.

### The Gas Loop

The first loop is pertaining to the gas where its journey starts from the gas supplier and traverse through multiple components resulting in power generation and retaining the gas byproduct. As part of this, the gas will be treated through all the intermediate stages as follows:

**Gas Supplier.** This component is essentially the source that is going to feed the gas into the system. Although the atmospheric air is the most reasonable and reachable type of gas, other options can be explored and utilized such as the emitted CO<sub>2</sub> from the system's components. All these gases are connected to the gas supplier through tubes and valves to ensure proper control. The next station of the gas will be the compressor.

**Energy Source.** Before diving into the compressor, it's worth discussing how this compressor, and the entire system, is powered up. Notwithstanding the possibility of utilizing a permanent power grid to energize the system, it can be more favorable to consider the option of renewable sources. This is because the nature of this idea revolves around desert environment where providing such power grid can be challenging.

**Compressor.** Generally, the function of the compressor in the system is to mechanically reduce the volume of the gas in which the molecules are packed more closely together, thereby increasing its pressure. There are many types of compressors, but this paper chooses the Centrifugal Compressor because it's the most appropriate for continuous operation and scalability of the system. When gas is compressed, its temperature rises significantly due to the adiabatic heating effect; a thermodynamic process that generates heat when the gas is compressed without heat exchange with the surroundings. This temperature increase can reduce the efficiency of the compression process and may pose material stress on the system components, and this is where the next component of the system is utilized.

**Heat Exchanger.** To counteract generated heat from the compressor, the system includes a heat exchanger that cools down the compressed gas by exchanging the heat with a coolant. Cooling the gas before it enters the underground storage vessels increases the density of the gas, allowing more energy to be stored in the same volume, thus improving the energy storage density of the system. This is further discussed in

the second loop; Coolant Loop which illustrates how the system utilizes the warm coolant in generating electricity.

***Underground Storage Vessel.*** The cooled gas existing heat exchanger is injected into the underground flexible vessel to store the energy in a potential form. The vessel, however, is designed to take advantage of the ambient heat from the surrounding desert sand to reheat the gas before it enters the turbine. This gas reheating increases the efficiency of the expansion process resulting in a more mechanical force, leading to a direct proportional power generation.

***Turbine Generator.*** When the energy is needed, the valve that controls the gas between the vessel and the turbine is triggered to open state resulting in gas flow that influence the turbine in generating electrical energy. As a result of this process, the gas formation produced is captured and fed back to the compressor. This closed-loop system minimizes gas loss and environmental impact, making the process sustainable and repeatable.

### **The Coolant Loop**

The second loop is for the coolant that is essential for efficiency maximization as well as to produce energy in low amount. The cold coolant starts its journey from the Coolant Supply going to the Heat Exchanger to cool down the hot gas. The coolant is then warmed up and stored into the Coolant Storage. This process resulting energy is fed into the turbine to produce electricity. The coolant continues its journey to the Coolant Supply.

The above process capitalizes on the solar energy as it relates to the sand heating, flexible vessel, and the system closed loops. The desert-based energy storage concept designed to help bridge the gap between when energy is available and when it is needed most. At its core, the system uses flexible vessels buried beneath desert sands, filled with a compressed working fluid during periods of surplus energy and discharged through a turbine to generate electricity during peak demand. The surrounding sand acts as a natural insulator, helping to minimize heat losses and maintain efficiency even in extreme climates.

One of the most promising aspects of this approach is its use of CO<sub>2</sub> or similar gases as the working fluid. This choice offers thermodynamic benefits for power generation while also turning a potential waste product into a valuable resource, supporting broader carbon reduction goals. Because it does not require large water reservoirs or specific geological formations, the system can be deployed in places where traditional storage technologies such as pumped hydro are simply not viable.

The flexibility of this design means it can store energy from both renewable and conventional sources, making it well suited for supporting solar and wind integration by providing backup power when conditions are less favorable. Its modular nature allows installations to start small and grow as demand or infrastructure evolves. By using underutilized desert land, it also minimizes land use conflicts and environmental impact.

This energy storage approach provides a practical, scalable, and sustainable solution to enhance grid reliability and accelerate the transition to cleaner energy sources. By leveraging the inherent advantages of desert environments and integrating engineered long-duration energy storage systems, it reduces reliance on oil-based power generation and facilitates higher penetration of renewable energy. For future work, a pilot projects based on this concept will play a crucial role in evaluating long-term system performance, real-world efficiency, and opportunities for cost optimization. If successfully demonstrated, this concept has the potential to serve as a replicable model for sustainable energy storage, not only in desert regions but also in arid and semi-arid areas worldwide

## **Environmental and Socio-Economic Impact**

The proposed underground compressed gas energy storage system presents significant environmental and socio-economic benefits, especially when deployed in arid and desert regions. From an environmental standpoint, this system offers a low-carbon alternative to conventional fossil fuel-based power generation, reducing greenhouse gas emissions and reliance on critical materials used in batteries. By leveraging

ambient desert heat for thermal enhancement, the technology eliminates the need for additional external thermal storage and reduces system complexity. Additionally, one major environmental benefit is the system's minimal surface disruption. Unlike pumped hydro or large-scale battery farms, which often require extensive land alteration, this system utilizes subterranean vessels that maintain the natural topography and biodiversity of the desert landscape. Franco (2025) emphasized the importance of low-footprint renewable energy infrastructure in sensitive ecological zones. Additionally, by avoiding the use of hazardous materials like lithium or cobalt, proposed system mitigates environmental contamination risks associated with mining and disposal of battery components.

From a water conservation perspective, the system is particularly suitable for arid climates where water resources are scarce. Conventional energy storage and generation systems often require substantial water for cooling or chemical processing. In contrast, this thermally passive approach conserves water and thus aligns with broader regional sustainability goals.

Socio-economically, the implementation of this system could serve as a catalyst for economic diversification and resilience in desert communities. According to Baniamerian et al. (2025), energy storage solutions that are adaptable to local environmental conditions can create job opportunities across the value chain—from system manufacturing and site preparation to maintenance and operation. Furthermore, the scalability and modularity of the system allow for decentralized deployment, which can be part of broader community-based energy initiatives.

This technology presents a viable route for local capacity building, knowledge transfer, and academic research and development. The strategic integration of such systems into national energy policies could stimulate public-private investment in green infrastructure and advance regional commitments to net-zero targets. Hunt et al. (2023) suggested that sustainable energy storage demonstrations in remote locations promote not just environmental stewardship but also community empowerment and long-term socio-economic stability. Peter (2025) emphasized that the co-location of pilot storage sites with academic institutions enhances the speed of innovation adoption and contributes to a skilled local workforce.

In the long term, the widespread adoption of this environmentally friendly and economically empowering storage system could serve as a replicable model for other arid and semi-arid zones worldwide. As highlighted by Hamiche et al. (2023), such systems can address both energy access and climate resilience simultaneously, making them highly attractive for international funding agencies and development banks.

## Conclusion

This paper presents a forward-thinking energy storage solution that draws on the principles of potential energy and natural thermal dynamics. By utilizing underground flexible vessels and ambient heat from desert environments, the system offers a scalable, efficient, and environmentally considerate method for storing and generating electricity potentially transforming how energy is managed in arid regions and beyond. The proposed thermodynamic model would measure the effectiveness of this solution that needs to be examined further as part of future work. Moreover, a proof of concept that would mimic the complete design, is to be explored as part of continuation of the patent innovation development phase. The primary focus will be on the flexible vessel material and the most effective approach on addressing the subterranean balance with the actual vessel design. Finally, the proposed thermodynamic model needs to be validated via either empirical or simulated data to ensure expected results are within the anticipated level of efficiency.

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